

TITLE:- Analysis of Concept Drift in Machine Learning-Based Network Intrusion Detection

1. INTRODUCTION

1.1 Background

Network Intrusion Detection Systems (NIDS) are used to identify malicious or abnormal activities in computer networks. Machine learning has become widely used in intrusion detection because it can learn patterns from network traffic and classify traffic as benign or malicious.

However, network traffic is not static. The characteristics of network traffic can change over time due to changes in users, applications, network behaviour, and attack patterns. When the statistical properties of data change, a machine learning model trained on earlier data may experience a decline in performance. This phenomenon is commonly associated with **concept drift and temporal distribution shift**.

Traditional machine learning evaluation often uses random train-test splitting. Although useful for measuring general classification performance, random splitting may mix traffic from different time periods and may not accurately represent how a model performs on future network traffic.

This project therefore investigates the performance of a machine learning-based intrusion detection model under **chronological changes in network traffic**. The study uses the CICIDS2017 dataset, evaluates a Random Forest model on later traffic periods, measures feature distribution changes using the Population Stability Index (PSI), and examines whether sequential model retraining can improve detection performance.

1.2 Motivation

A model that performs well on historical test data may not necessarily maintain the same performance when deployed on future network traffic.

Therefore, this project focuses on the following practical question:

Q.How does temporal change in network traffic affect the performance of a machine learning-based intrusion detection system, and can model adaptation reduce this performance degradation?

2. PROBLEM STATEMENT

Machine learning-based intrusion detection models are commonly evaluated using randomly divided datasets. Such evaluation may not adequately represent real-world deployment, where models encounter network traffic generated at later points in time.

Changes in traffic distributions and the appearance of new attack patterns can reduce the ability of a trained model to detect attacks accurately.

Therefore, there is a need to evaluate intrusion detection models using **chronological data**, identify changes in network traffic distributions, and examine whether periodic model adaptation can maintain detection performance.

3. OBJECTIVES

The main objectives of this project are:

1. To develop a machine learning-based network intrusion detection model using the CICIDS2017 dataset.
2. To evaluate the model using a **chronological train-test setup** rather than relying only on random splitting.
3. To measure changes in network traffic feature distributions across different time periods using **Population Stability Index (PSI)**.
4. To analyze the effect of temporal changes on intrusion detection performance using Accuracy, Precision, Recall, and F1-score.
5. To evaluate whether **sequential model retraining** can reduce performance degradation on later network traffic.

4. SCOPE OF THE PROJECT

The project focuses on:

- Binary classification of network traffic into **Benign** and **Attack**.
- CICIDS2017 network traffic data.
- Random Forest as the primary machine learning model.
- Chronological evaluation across different traffic periods.
- Distribution-shift analysis using PSI.
- Sequential retraining using previously observed labelled traffic.
- Evaluation using Accuracy, Precision, Recall, F1-score, and confusion matrices.

The project does not attempt to develop a production-level real-time intrusion detection system.

5. LITERATURE REVIEW

5.1 Concept Drift in Machine Learning

Concept drift refers to changes in the underlying data distribution over time that can affect the performance of a previously trained machine learning model. Gama et al. [1] provide a comprehensive survey of concept drift adaptation methods and discuss approaches for detecting and adapting to changing data. Lu et al. [2] further describe concept drift as a change in the statistical properties of data over time and categorize the problem into drift detection, understanding, and adaptation. These studies establish the importance of evaluating machine learning models when the data distribution changes.

5.2 Machine Learning for Network Intrusion Detection

Machine learning is widely used in network intrusion detection to identify patterns associated with benign and malicious network traffic. Random Forest, introduced by Breiman [3], is an ensemble learning method based on multiple decision trees and is suitable for classification tasks involving a large number of features. In this project, Random Forest is used as the primary classifier for binary intrusion detection.

5.3 Network Intrusion Detection Datasets

The quality and characteristics of an intrusion detection dataset have a significant effect on model evaluation. Ring et al. [4] reviewed network-based intrusion detection datasets and highlighted differences in traffic

characteristics, attack scenarios, data volume, and evaluation suitability. Their work emphasizes the importance of selecting an appropriate dataset when studying network intrusion detection.

The CICIDS2017 dataset was developed to provide realistic benign and attack traffic and contains multiple attack scenarios collected across five days. The dataset includes labeled network flows and more than 80 extracted network-flow features, making it suitable for machine learning-based intrusion detection research [5].

5.4 Research Gap

The reviewed literature establishes the importance of concept drift, machine learning-based intrusion detection, and suitable evaluation datasets. However, a model evaluated only through a random train-test split may not reflect its performance on later network traffic because temporal order is not preserved.

Therefore, this project focuses on chronological evaluation of CICIDS2017 traffic. It combines performance evaluation across later traffic periods, feature distribution analysis using the Population Stability Index (PSI), and sequential model retraining to examine whether model adaptation can reduce performance degradation.

6. DATASET DESCRIPTION

6.1 Dataset Used

The project uses the **CICIDS2017 (Canadian Institute for Cybersecurity Intrusion Detection System 2017)** dataset. It contains network traffic collected over multiple days and includes both **benign traffic and various types of network attacks**.

The dataset provides flow-based network features such as packet counts, packet lengths, flow duration, inter-arrival times, TCP flags, and other traffic characteristics.

6.2 Dataset Structure

The downloaded Machine Learning CSV files contain **8 traffic files** corresponding to different days and attack scenarios:

Day	Main Traffic / Attack Types
Monday	Benign
Tuesday	FTP-Patator, SSH-Patator
Wednesday	DoS attacks, Heartbleed
Thursday	Web Attacks, Infiltration
Friday	Bot, PortScan, DDoS

For this project, **Monday was excluded from model training** because it contains only benign traffic and no attack samples.

6.3 Target Variable

The original `Label` column contains multiple attack categories. For this project, these labels were converted into a binary classification:

- **0 → BENIGN**
- **1 → ATTACK**

The original attack labels were retained separately for reference.

6.4 Data Preprocessing

The following preprocessing steps were performed:

1. Column names and labels were standardized.
2. Infinite values were converted to missing values.
3. Missing numerical values were replaced using **median imputation**.
4. Duplicate records were retained rather than removed blindly.
5. The original labels were preserved.
6. Only the **78 network traffic features** were used for model training.

6.5 Dataset Distribution

After combining Tuesday–Friday data used for the main experiments:

- **Total records:** 2,300,825
- **Benign:** 1,743,179
- **Attack:** 557,646
- **Features used:** 78

The dataset was evaluated chronologically to study how model performance changes when it encounters traffic from later periods.

Dataset Source: Canadian Institute for Cybersecurity, University of New Brunswick (UNB).

Link:- <https://www.unb.ca/cic/datasets/ids-2017.html> for database.

7. METHODOLOGY

The project follows a chronological machine learning approach to study the effect of temporal changes in network traffic on intrusion detection performance.

7.1 Data Preprocessing

The CICIDS2017 data was cleaned and prepared before model training. Infinite values were converted to missing values and numerical missing values were handled using median imputation. The original attack labels were retained, while a binary target variable was created:

- **0 – Benign**
- **1 – Attack**

A total of **78 network traffic features** were used for model training.

7.2 Model Development

A **Random Forest Classifier** was selected as the primary machine learning model. Random Forest was chosen because it can handle a large number of numerical features and is suitable for classification problems involving complex traffic patterns.

The model was trained using traffic from **Tuesday and Wednesday**, while Monday was excluded because it contained only benign traffic.

7.3 Baseline Evaluation

A stratified random sample of **500,000 records** was used to establish a baseline performance.

The data was divided into:

- **80% training data**
- **20% testing data**

This experiment provides a conventional random-split performance reference.

7.4 Chronological Evaluation

To simulate real-world deployment, the model was trained using Tuesday and Wednesday traffic and then evaluated on later periods in chronological order:

Thursday WebAttacks → Thursday Infiltration → Friday Morning → Friday PortScan → Friday DDoS

The model was kept **static** during this experiment, meaning it was not retrained between evaluation periods.

Performance was measured using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

7.5 Distribution Shift Analysis

The **Population Stability Index (PSI)** was used to measure changes in the distribution of network traffic features between the initial training period and each later evaluation period.

The PSI values were used as an indicator of temporal distribution shift.

7.6 Sequential Model Adaptation

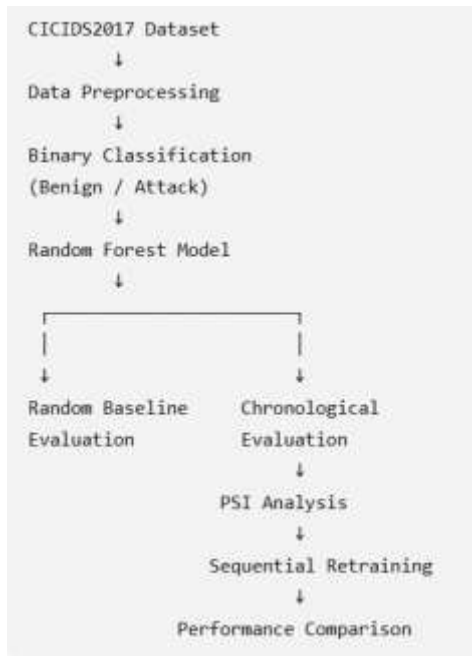
A second experiment evaluated whether periodic retraining could help the model adapt to changing traffic.

The process was:

Initial Training → Evaluate Thursday → Update Model → Evaluate Infiltration → Update Model → Evaluate Friday Morning → Update Model → Evaluate PortScan → Update Model → Final DDoS Evaluation

Only traffic from a **completed evaluation period** was used for subsequent model updates, preventing future evaluation data from being used before its evaluation.

7.7 Overall Experimental Flow



8. SYSTEM / EXPERIMENTAL DESIGN

8.1 Experimental Setup

The experiments were performed using **Python** in Google Colab. Pandas and NumPy were used for data processing, Scikit-learn was used for machine learning, and Matplotlib was used for visualization.

Hardware/Software Environment:

- Programming Language: Python
- Environment: Google Colab
- Data Processing: Pandas, NumPy
- Machine Learning: Scikit-learn
- Visualization: Matplotlib
- Dataset: CICIDS2017
- Primary Model: Random Forest Classifier

8.2 Experimental Phases

The project was divided into four main phases:

1. Baseline Evaluation

Random train-test splitting was used to obtain a conventional performance reference.

2. Chronological Evaluation

A model trained on earlier traffic was evaluated on later traffic periods without retraining.

3. Distribution Shift Analysis

PSI was calculated to identify changes in feature distributions between the training period and later periods.

4. Sequential Model Adaptation

The model was periodically retrained using labelled data from completed traffic periods and evaluated on the next unseen period.

8.3 Evaluation Metrics

The following metrics were used:

- **Accuracy:** Overall proportion of correctly classified traffic.
- **Precision:** Proportion of predicted attacks that were actually attacks.
- **Recall:** Proportion of actual attacks correctly detected.
- **F1-score:** Harmonic mean of precision and recall.
- **Confusion Matrix:** Shows true and false classifications for benign and attack traffic.

9. RESULTS AND DISCUSSION

9.1 Baseline Random Forest Performance

The Random Forest model was first evaluated using a stratified random train-test split of 500,000 records.

Metric	Score
Accuracy	99.84%
Precision	99.69%
Recall	99.63%
F1-Score	99.66%

The high baseline performance shows that the model can effectively classify benign and attack traffic when training and testing data are randomly distributed.

However, this evaluation does not represent a strictly future-data scenario because traffic from different periods can be present in both training and testing sets.

9.2 Chronological Evaluation

The model was then trained using Tuesday and Wednesday traffic and evaluated on subsequent traffic periods without retraining.

Evaluation Period	Accuracy	Precision	Recall	F1-Score
Thursday WebAttacks	98.71%	42.86%	2.06%	3.94%
Thursday Infiltration	99.94%	0%	0%	0%
Friday Morning	98.94%	0%	0%	0%
Friday PortScan	44.56%	76.19%	0.09%	0.18%
Friday DDoS	79.10%	99.96%	63.18%	77.43%

The chronological results show substantial variation in model performance across later traffic periods. In particular, attack-class recall and F1-score decreased considerably for several periods despite high accuracy.

For example, the model achieved **99.94% accuracy on Thursday Infiltration but detected none of the attack instances**, resulting in a recall and F1-score of 0%.

This demonstrates why accuracy alone is insufficient for evaluating an intrusion detection system, particularly when the data is imbalanced.

9.3 Distribution Shift Analysis Using PSI

PSI was calculated between the initial training period and each later evaluation period.

Period	Mean PSI	Features with Substantial Shift
Thursday WebAttacks	0.1633	19
Thursday Infiltration	0.2273	33
Friday Morning	0.1638	21

Period	Mean PSI	Features with Substantial Shift
Friday PortScan	0.6553	51
Friday DDoS	0.4465	47

The results indicate measurable changes in network traffic feature distributions across all evaluation periods.

Friday PortScan showed the largest distribution shift, with a mean PSI of **0.6553** and 51 of the 78 features showing substantial shift. This period also produced very low attack recall and F1-score in the static model.

PSI indicates distributional change, but it does not by itself prove concept drift. Performance degradation may also be influenced by the presence of attack types that were not present in the initial training period.

9.4 Sequential Model Adaptation

The model was sequentially retrained using labelled data from previously completed periods.

Period	Static F1	Adapted F1
Thursday WebAttacks	3.94%	3.94%
Thursday Infiltration	0%	0%
Friday Morning	0%	0%
Friday PortScan	0.18%	0.19%
Friday DDoS	77.43%	77.33%

The sequential retraining strategy produced only small changes in performance in this experiment. It did not substantially recover the low attack-detection performance observed for WebAttacks, Infiltration, Friday Morning, or PortScan.

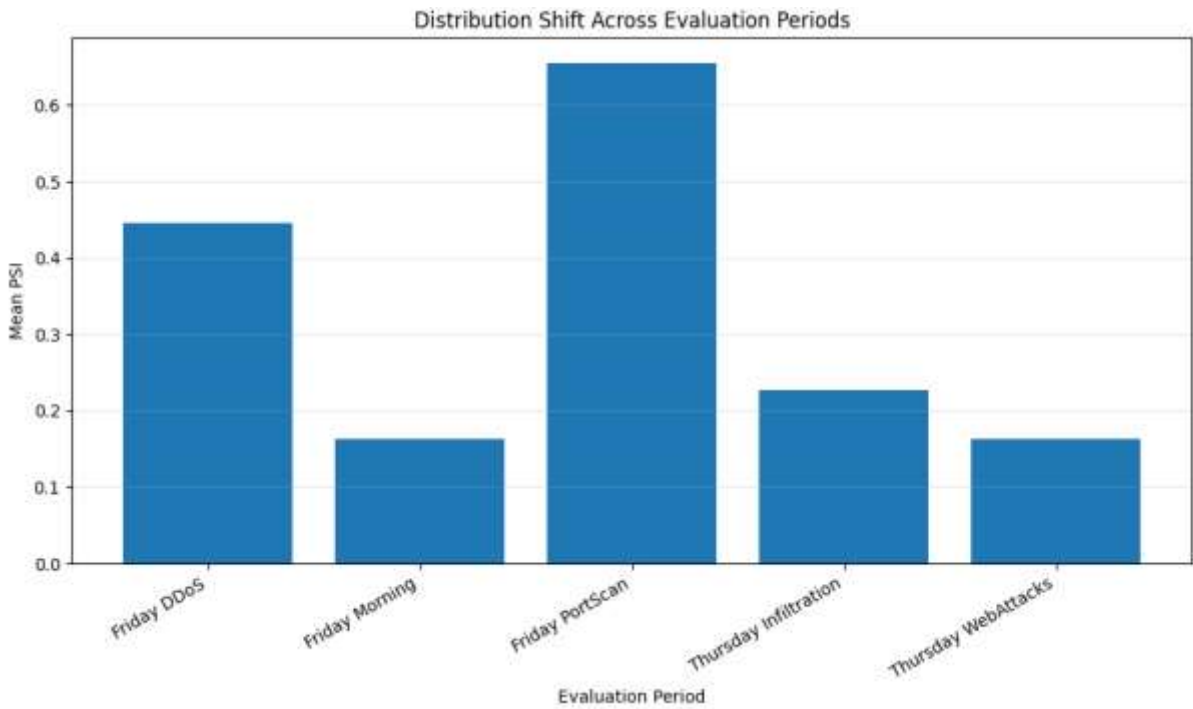
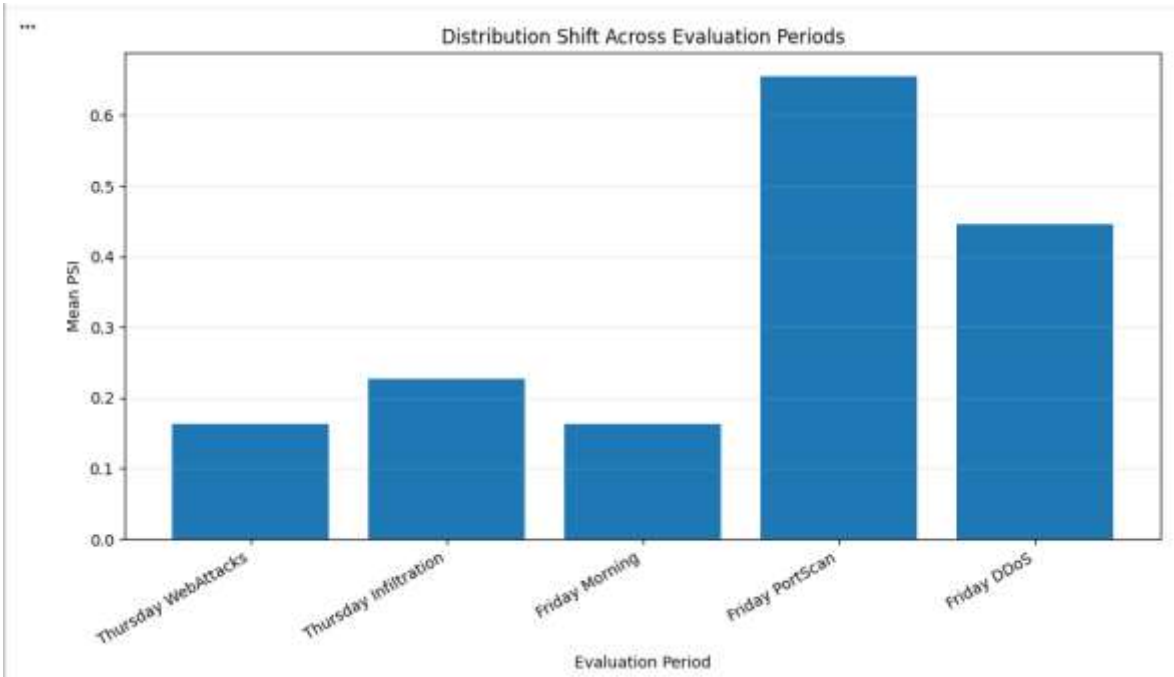
This suggests that **simple periodic retraining using sampled labelled data was not sufficient to overcome the observed changes in this particular experimental setup.**

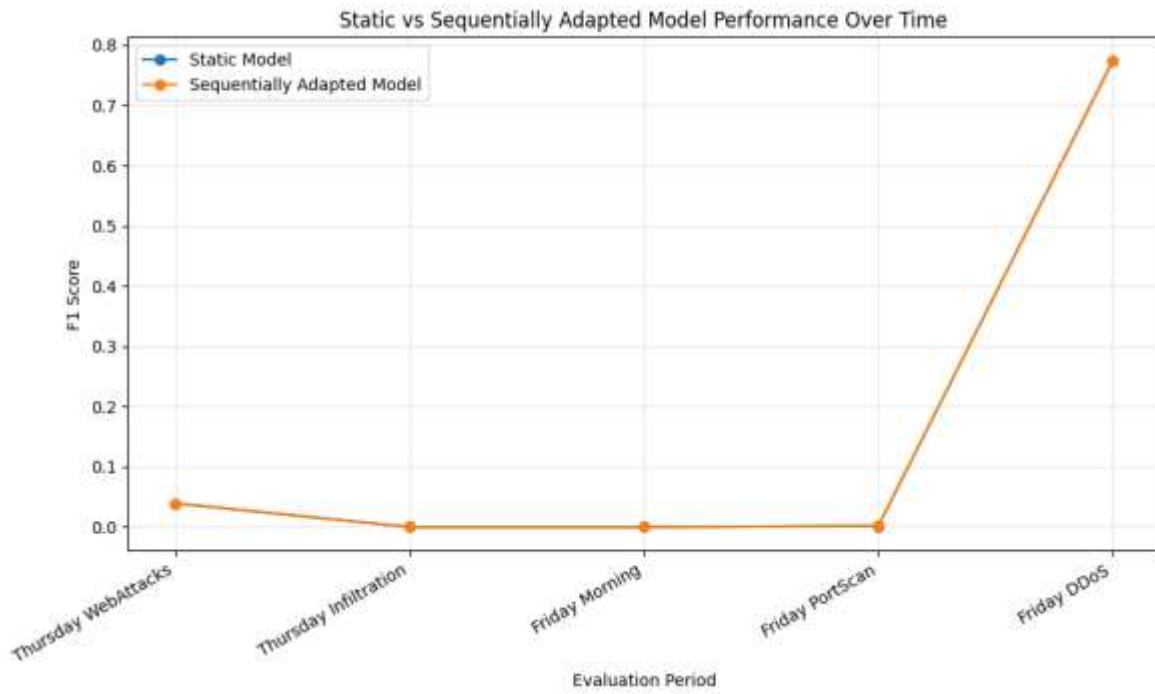
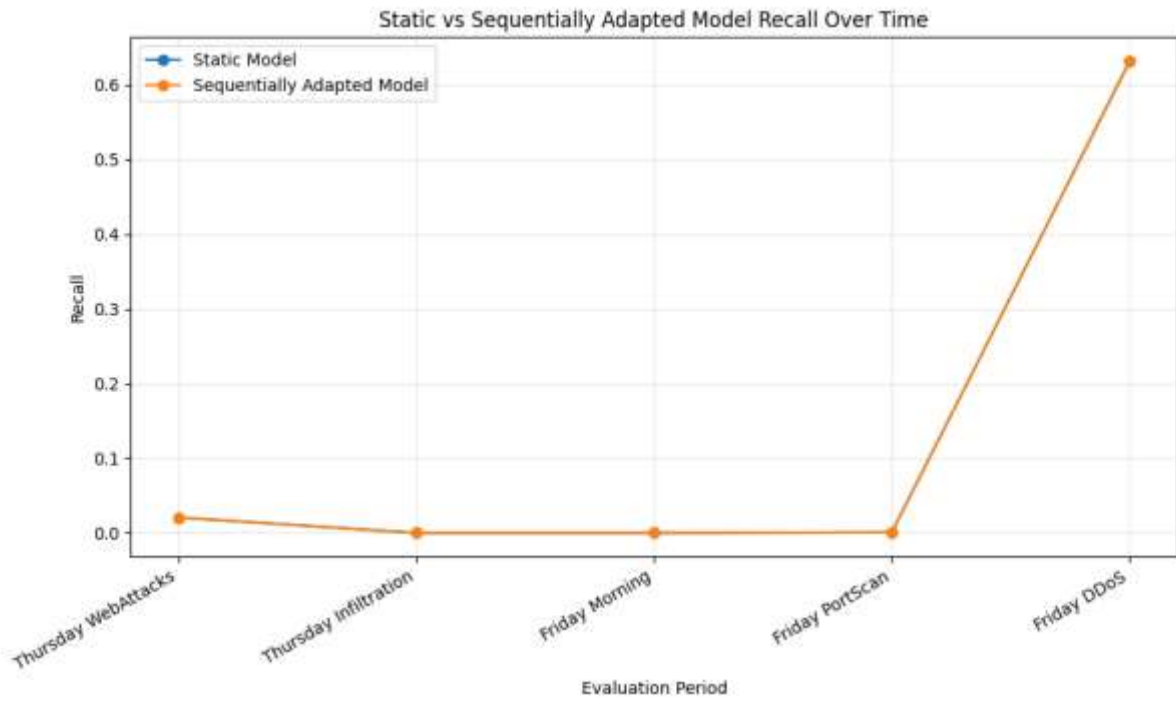
9.5 Overall Discussion

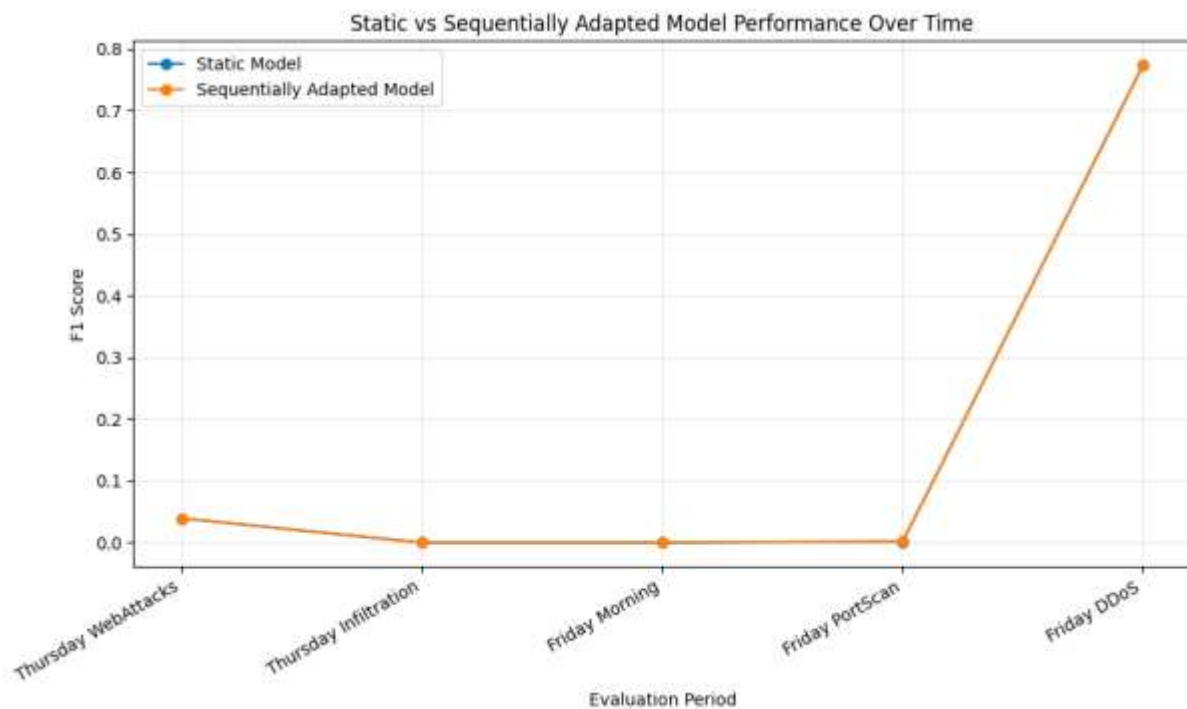
The experiments demonstrate that a Random Forest model can achieve very high performance under conventional random evaluation but can show substantially different performance when evaluated chronologically on later network traffic.

The PSI analysis confirms that the statistical distribution of network features changed across the evaluation periods. However, the relationship between distribution shift and model performance was not uniform across all periods.

Overall, the findings are **consistent with temporal distribution shift** and demonstrate the importance of evaluating intrusion detection models using future-like chronological data rather than relying only on random train-test splits.







10. LIMITATIONS

The following limitations were considered during the project:

1. **No row-level timestamps:** The Machine Learning CSV files used from CICIDS2017 do not contain a Timestamp column. Therefore, temporal ordering was based on the documented capture-day and traffic-session sequence.
2. **Limited model selection:** The experiments primarily focused on the **Random Forest** classifier. Other machine learning algorithms were not included in the final comparative analysis.
3. **New attack types:** Some later evaluation periods contain attack types that were not present in the initial Tuesday–Wednesday training data. Therefore, performance degradation may be influenced by both **distribution shift and previously unseen attack patterns**.
4. **PSI interpretation:** PSI was used as an indicator of feature distribution change. A high PSI value alone does not establish the presence of concept drift.
5. **Label availability for adaptation:** The sequential retraining experiment assumes that labelled data from a completed traffic period becomes available before the model is updated.
6. **Dataset-specific evaluation:** The findings are based on CICIDS2017 and may not directly represent the behaviour of intrusion detection models on other network environments or datasets.

11. FUTURE SCOPE

The project can be extended in the following ways:

1. Real-Time Drift Detection

Implement automated monitoring to detect changes in incoming network traffic distributions.

2. Adaptive Machine Learning Models

Explore online learning and incremental learning techniques that can continuously adapt to changing traffic patterns.

3. Real-Time Intrusion Detection System

Integrate the trained model with live network traffic to develop a real-time intrusion detection system.

4. Automated Model Retraining

Develop an automated pipeline that triggers model retraining when significant distribution changes or performance degradation are detected.

5. Comparison of Multiple Models

Future work can compare Random Forest with algorithms such as XGBoost, SVM, neural networks, and other adaptive learning approaches.

6. Evaluation on Recent Datasets

The methodology can be tested on newer intrusion-detection datasets to determine whether the observations remain consistent across different network environments.

7. Interactive Monitoring Dashboard

A dashboard can be developed to display traffic statistics, PSI values, model performance, detected attacks, and drift alerts in real time.

In short:

The project can be extended from an offline analysis of concept drift into an adaptive, real-time intrusion detection system capable of monitoring and responding to changing network behaviour.

12. CONCLUSION

This project analyzed the effect of temporal changes in network traffic on the performance of a machine learning-based network intrusion detection model using the CICIDS2017 dataset.

The Random Forest model achieved high performance under conventional random train-test evaluation, with an F1-score of 99.66%. However, chronological evaluation on later traffic periods showed substantial variations in attack detection performance, particularly in Recall and F1-score. This demonstrates that high performance under random evaluation does not necessarily represent performance on future network traffic.

The PSI analysis identified measurable changes in network traffic feature distributions across the evaluation periods. The largest distribution shift was observed for the Friday PortScan traffic, which also showed very low attack detection performance.

Sequential retraining was evaluated as a simple adaptation strategy. In this experimental setup, retraining produced only minor changes in performance and did not substantially recover the degradation observed for several attack periods.

Overall, the study highlights the importance of chronological evaluation and monitoring of data distribution changes when developing machine learning-based intrusion detection systems. The findings are consistent with temporal distribution shift and show why models deployed in changing network environments may require continuous monitoring and more advanced adaptation strategies.

13. REFERENCES

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